

Problem Set 5
Winter 2020

Issued: Thursday, February 13

Due: Monday, February 24

Full credit: 110 points.

Consider the canonical d -dimensional exponential family with densities (w.r.t. ν)

$$p_\theta(x) = \exp(\langle \theta, T(x) \rangle - A(\theta)) h(x), \quad \theta \in \Omega$$

where $\Omega \subset \Omega_0 = \{\theta : A(\theta) < \infty\}$. Here, Ω_0 is the natural parameter space. The family is called **full** if $\Omega = \Omega_0$.

Problem 5.1 (25 points)

Let $Z \in \mathbb{R}^{n \times p}$ be a fixed deterministic matrix and $Y \sim N(Z\beta, I_n)$, $\beta \in \mathbb{R}^p$. Here, I_n is the $n \times n$ identity matrix.

- (a) Show that the distribution of Y can be represented as an **n -dimensional** canonical exponential family and specify A , T and both Ω and Ω_0 , its natural parameter space. Is this a full family?
- (b) Under what condition on Z , the above family is full-rank?
- (c) Show that the distribution of Y can be represented as an **p -dimensional** canonical exponential family and specify A , T and both Ω and Ω_0 , its natural parameter space.
- (d) Under what condition on Z , the preceding representation is full-rank? Assuming that this condition holds, derive ∇A and ∇A^{-1} (i.e., the inverse of the gradient map).

Problem 5.2 (20 points)

(This problem requires some basic understanding of measure theory.)

Consider a standard exponential family with densities $p_\theta(x) = \exp(\langle \theta, x \rangle - A(\theta))$ w.r.t. ν , where both $x, \theta \in \mathbb{R}^d$. Assume that $\{p_\theta\}$ is a model for random vector X . Let $B \in \mathbb{R}^{d \times d}$ be a non-singular matrix and define

$$Z = BX + z_0, \quad \phi = B^{-T}\theta + \phi_0 \tag{1}$$

where $B^{-T} = (B^{-1})^T = (B^T)^{-1}$.

- (a) Show that the distributions for Z form a standard d -dimensional exponential family with parameter ϕ , w.r.t. to the measure ν_1 defined by

$$\nu_1(dz) = e^{-\langle \phi_0, z \rangle} \nu_2(dz), \quad \text{where } \nu_2(A) = \nu(B^{-1}(A - z_0))$$

for any event $\mathcal{A}$. Also argue that this new family is equivalent to the original family. (Try to define what equivalence means in this context yourself).

- (b) Let Ω_0 and Γ_0 be the natural parameter spaces of the families associated with X and Z . Moreover, let μ be a given vector in the interior of the set of realizable means for measure ν (i.e., $\mu = \int x p(x) \nu(dx)$ for some density p). Choose the transformation in (1) such that 0 belongs to the interior of the set of realizable means for measure ν_1 . Argue that for that transformation, if there is $\phi_1 \in \text{int}(\Gamma_0)$ such that $\mathbb{E}_{\phi_1}[Z] = 0$, then there is $\theta_1 \in \text{int}(\Omega_0)$ such that $\mathbb{E}_{\theta_1}[X] = \mu$.

Problem 5.3 (20 points)

Let $X_1, \dots, X_n$ be iid $\text{Poi}(\theta)$ and let $\bar{x}$ be the observed average of $\{X_i\}$. Assume that we put a $\text{Gamma}(a, b)$ prior on Θ (i.e., $\pi(\theta) \propto \theta^{a-1} e^{-\theta/b}$).

- (a) Show that the posterior on Θ is $\text{Gamma}(a + n\bar{x}, b/(1 + nb))$.
(b) Derive the Bayes rule under the loss $L(\theta, \delta) = (\theta - \delta)^2 / \theta^k$ for some $k > 0$. Assume $a - k > 0$.

Solution 5.1

For part (b), assuming $a - k > 0$,

$$\delta^k(\bar{x}) = \frac{\mathbb{E}(\theta^{1-k} | \bar{x})}{\mathbb{E}(\theta^{-k} | \bar{x})} = \frac{b}{1 + nb} (n\bar{x} + a - k).$$

Problem 5.4 (20 points)

Consider a one-way random effects model that can be written as the hierarchical Bayesian model

$$\begin{aligned} X_{ij} | \mu, \alpha_i &\sim N(\mu + \alpha_i, \sigma^2), \quad j = 1, \dots, n, \quad i = 1, \dots, s, \\ \alpha_i &\sim N(0, \sigma_A^2), \quad i = 1, \dots, s \end{aligned}$$

If, in addition, we specify that $\mu \sim \text{Uniform}(-\infty, \infty)$, an improper prior, derive the Bayes estimator of $\mu + \alpha_i$ under squared error loss.

Hint: $\text{Uniform}(-\infty, \infty)$ is an *improper* prior, basically, the Lebesgue measure on $\mathbb{R}$ (or any constant multiple of it). It has constant density w.r.t. Lebesgue measure, but it cannot be normalized to have total mass 1 over $\mathbb{R}$. Still as long as the posterior integrates to 1, we can pretend everything is fine. Just go through the calculations setting the prior to $\pi(\theta) = 1$ for all $\theta \in \mathbb{R}$.

Problem 5.5 (20 points)

Consider the i.i.d. linear observation model

$$Y_i = x_i^T \beta + W_i, \quad i = 1, \dots, n$$

where $\beta \in \mathbb{R}^p$, the design vectors $x_i \in \mathbb{R}^p$ are fixed and known, and $W_i \sim N(0, \sigma^2)$ is observation noise. Assume that $p < n$.

- (a) Assume that $\sigma^2 > 0$ is known, and that β is modeled as a fixed but unknown vector. Find the least-squares estimate of β based on Y .

- (b) Now assume that β is distributed according to the prior $N(\beta_0, \sigma^2 Q)$, where $Q \in \mathbb{R}^{p \times p}$ is a known, positive definite symmetric matrix. Find the Bayes estimator of $z^T \beta$ under least-squares loss where $z \in \mathbb{R}^p$ is a known fixed vector.
- (c) Suppose that the minimum eigenvalue of $\sum_{i=1}^n x_i x_i^T$ tends to ∞ . Show that the estimator from part (b) is consistent (converges in probability to the correct β) as $n \rightarrow +\infty$.

Solution 5.2

- (a) Let X be an $n \times p$ matrix with x_i^T as row i , for $i = 1, \dots, n$. The least-squares estimate of β based on Y minimizes $\|Y - X\beta\|^2$. Solving yields that

$$\hat{\beta} = (X^T X)^{-1} X^T Y = \beta^* + (X^T X)^{-1} X^T W$$

- (b) The joint distribution of (Y, β) —given by product of the prior on β and the conditional distribution of Y given β —is proportional to

$$\exp \left\{ -\|Y - X\beta\|_2^2 / (2\sigma^2) \right\} \times \exp \left\{ (\beta - \beta_0)^T Q^{-1} (\beta - \beta_0) / (2\sigma^2) \right\}.$$

Define $\tilde{\beta} = (X^T X + Q^{-1})^{-1} (Q^{-1} \beta_0 + X^T X \hat{\beta})$. Some algebra yields that the joint over (Y, β) is proportional to

$$\exp \left\{ -\frac{(\beta - \tilde{\beta})^T (X^T X + Q^{-1}) (\beta - \tilde{\beta})}{2\sigma^2} \right\},$$

so that the posterior distribution of β is given by $N(\tilde{\beta}, \sigma^2 (X^T X + Q^{-1})^{-1})$. Thus, the Bayes least squares estimate of $z^T \beta$ is simply $z^T \tilde{\beta}$.

- (c) If the minimum eigenvalue of $X^T X$ tends to ∞ , then $(X^T X + Q^{-1})^{-1} Q^{-1} \beta_0$ converges to 0, and $\tilde{\beta}$ converges to β^* . Moreover, this implies that $(X^T X + Q^{-1})^{-1} X^T X \hat{\beta}$ converges to β^* , whence $\tilde{\beta}$ converges to β^* as well. Thus, $z^T \tilde{\beta}$ converges to $z^T \beta^*$. (After diagonalization of $X^T X$ and applying WLLN, you can show above things.)

Problem 5.6 (20 points)

Let $(X_1, \dots, X_n)$ be an i.i.d. sample from the uniform distribution on $(0, \theta)$, where $\theta > 0$ is unknown. Suppose that the prior distribution of θ is log-normal with parameters (μ_0, σ_0^2) where $\mu_0 \in \mathbb{R}$ and $\sigma_0^2 > 0$ are known constants.

- (a) Find the posterior density of $\log \theta$.
- (b) Suppose that we are interested in estimating θ under the loss function

$$L(\delta, \theta) = \begin{cases} 0 & \text{if } \delta = \theta \\ 1 & \text{otherwise.} \end{cases}$$

Find the Bayes estimator of θ under this loss function.

(Hint 1: Part (a) is related.)

(Hint 2: With 0-1 loss the Bayes rule would be the Maximum A Posteriori (MAP) estimator, i.e., the one maximizing $\mathbb{P}(\Theta = \delta|X)$ over δ . This makes sense for discrete variables. For a continuous posterior having density $\pi(\theta|x)$ w.r.t. Lebesgue measure, i.e., $\mathbb{P}(\Theta \in A|X) = \int_A \pi(\theta|x)dx$, we interpret MAP as maximizing $\mathbb{P}(\Theta \in (\delta, \delta + d\delta)|X) \approx \pi(\delta|X)d\delta$ over δ . Thus, MAP in this case is the *mode* of the posterior.

Problem 5.7 (15 points)

Given a family $\{p_\theta(x) \mid \theta \in \Omega\}$ and an estimator δ with $g(\theta) = \mathbb{E}_\theta[\delta(X)]$, the information bound is

$$B(\theta) = [g'(\theta)]^2 / I(\theta).$$

Now suppose that we reparameterize the family in terms of $\eta = h(\theta)$, where h is differentiable and invertible. That is, consider the reparametrized family

$$\{\tilde{p}_\eta(x) = p_{h^{-1}(\eta)}(x) \mid \eta \in h(\Omega)\}$$

and let δ be an unbiased estimator of $\tilde{g}(\eta) = g(h^{-1}(\eta))$.

Show that the Fisher information $\tilde{I}(\eta)$ is *not* invariant, but that the information bound $\tilde{B}(\eta)$ is invariant to this reparameterization.

Solution 5.3

Recall that $\ell_\theta(X) = \log p_\theta(X)$. It helps to use the notation $\eta = \eta(\theta)$ and $\theta = \theta(\eta)$, instead of $\eta = h(\theta)$ and $\theta = h^{-1}(\eta)$. We can write

$$\frac{\partial}{\partial \eta} \ell_{\theta(\eta)}(X) = \theta'(\eta) \dot{\ell}_{\theta(\eta)}(X) \implies \tilde{I}(\eta) = \mathbb{E}_{\theta(\eta)} \left[\frac{\partial}{\partial \eta} \ell_{\theta(\eta)}(X) \right]^2 = [\theta'(\eta)]^2 I(\theta(\eta)).$$

Reversing the role of η and θ , we have by symmetry $I(\theta) = [\eta'(\theta)]^2 \tilde{I}(\eta(\theta))$ (This can also be seen by the inverse function theorem.) We observe that Fisher information is not invariant.

Similarly $\tilde{g}'(\eta) = \frac{\partial}{\partial \eta} g(\theta(\eta)) = \theta'(\eta) g'(\theta(\eta))$. We obtain

$$\widetilde{\text{CRB}}(\eta) = \frac{[\tilde{g}'(\eta)]^2}{\tilde{I}(\eta)} = \frac{[\theta'(\eta) g'(\theta(\eta))]^2}{[\theta'(\eta)]^2 I(\theta(\eta))} = \frac{[g'(\theta)]^2}{I(\theta)} = \text{CRB}(\theta)$$

showing that CRB is indeed invariant to reparametrization.

Problem 5.8 (10 points)

Let $B \succ 0$ be a positive definite $d \times d$ matrix, and $v \in \mathbb{R}^d$. Prove that

$$\sup_{x \neq 0} \frac{(x^T v)^2}{x^T B x} = v^T B^{-1} v.$$

This result is useful when proving the multidimensional version of information inequality. (Hint: Change of variable $z = B^{1/2}x$ might be helpful.)

Solution 5.4

By the change of variable $z = B^{1/2}x$, we have

$$\sup_{x \neq 0} \frac{(x^T v)^2}{x^T B x} = \sup_{z \neq 0} \frac{(z^T B^{-1/2} v)^2}{z^T z} = \|B^{-1/2} v\|_2^2 = v^T B^{-1} v.$$

where we have used the Cauchy-Schwarz inequality, alternatively written as

$$\sup_{z \neq 0} \frac{(z^T x)^2}{z^T z} = \sup_{\|u\|_2=1} (u^T x)^2 = \|x\|_2^2$$

where $u = z/\|z\|_2$.